BİL 455/533 – Parallel Processing

Exercise #1

1. **Multi-threaded histogram computation.** You can use the OpenCV's C++ API to read-in images and access pixel values. You can use the provided image lena_grayscale_hq.jpg.
   i. Write a single-threaded C++ code that computes the histograms of 8-bit grayscale images.
   ii. Write a multi-threaded C++ code that computes the histograms of 8-bit grayscale images. **Use the standard C++ threads and atomic data types. Do not use mutexes**.
   iii. Write a multi-threaded C++ code that computes the histograms of 8-bit grayscale images. **Use the standard C++ threads and mutexes. Do not use atomic data types.**
   iv. Compare the execution times of the codes in parts (i), (ii) and (iii) for varying numbers of threads, for example, 4-threads, 8-threads, maximum number of threads your system allows.

2. **Multi-threaded integral array computation.** You can assume 32-bit int data type.
   i. Write a single-threaded C++ code that computes the integral arrays of int arrays.
   ii. Write a multi-threaded C++ code that computes the integral arrays of int arrays. Use the standard C++ threads.
   iii. Assuming a fixed array length of 4096 compare the execution times of the codes in parts (i) and (ii) for varying numbers of threads, for example, 4-threads, 8-threads, maximum number of threads your system allows.
   iv. Compare the execution times of the codes in parts (i) and (ii) for varying array sizes, for example, 4096 integers, 4 million integers, maximum size your system allows.

3. **Multi-threaded integral image computation.** You can use the OpenCV's C++ API to read-in images and access pixel values. You can use the provided image lena_grayscale_hq.jpg.
   i. Write a single-threaded C++ code that computes the integral images of 8-bit grayscale images.
   ii. Write a multi-threaded C++ code that computes the integral images of 8-bit grayscale images. Use the standard C++ threads.
   iii. Assuming a fixed image size of 100x100 compare the execution times of the codes in parts (i) and (ii) for varying numbers of threads, for example, 4-threads, 8-threads, maximum number of threads your system allows. You can use OpenCV's C++ API to resize your image to 100x100.
   iv. Compare the execution times of the codes in parts (i) and (ii) for varying image sizes. You can use OpenCV's C++ API to resize your image to obtain "more" pixels…